

Sobre la acción del grupo de automorfismos de una superficie de Riemann compacta en el conjunto de sus sístoles

Camila Guajardo Vásquez

Defensa de Tesis de Magíster
Directora: Dra. Anita Rojas Rodríguez

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- ▶ **Systoles**: non-homotopically trivial simple closed geodesics of **minimal length**.
- ▶ **Systolic length (sys)**: length of any systole.
- ▶ **Kissing number (Kiss)**: number of systoles, up to free homotopy.
- ▶ **Purpose**: Provide bounds on the systolic length and the kissing number.

Introduction

Method: Computational approach.

- ▶ Input: abstract **finite group** and a **signature**.
- ▶ Output: Fundamental polygon for a surface with that group action
- ▶ **Reference:** Antonio Behn, Rubí Rodríguez and Anita Rojas (2013).
- ▶ Length of selected closed geodesics bounds sys
- ▶ **Group action by isometries** gives a bound on Kiss.

Guiding example: **Bolza surface**, endowed with an action of $GL_2(3)$ with signature $(0; 8, 3, 2)$. Test case for the algorithm.

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Bolza surface

- Fundamental polygon

- Homology basis and group action

Families on $\mathcal{M}_2$

- Parametrizing polygons

- Systolic bounds

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Fundamental polygons

Unit disc model: easy to visualize, concrete geometry.

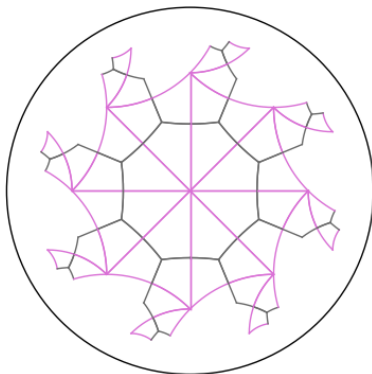


Figure: Fundamental polygon for the uniformizing Fuchsian group Γ for the Bolza surface.

Geometry in $\mathbb{D}$ induce geometry in B

- ▶ **Metric:** $g_z = \frac{4}{(1 - |z|^2)^2} dzd\bar{z}$.
- ▶ **Isometries:** $\text{Aut}(\mathbb{D}) = \left\{ \begin{pmatrix} e^{i\theta} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -a \\ -\bar{a} & 1 \end{pmatrix} : \theta \in [0, 2\pi], a \in \mathbb{D} \right\}$

For the Bolza surface B :

- ▶ **Induced metric:** Restrict to a fundamental polygon.
- ▶ $\text{Aut}(B) = N(\Gamma)/\Gamma$ implies **$\text{Aut}(B)$ are isometries.**
- ▶ **Closed geodesics** are geodesics contained in the fundamental polygon, endpoints are Γ -identified.

Closed geodesic

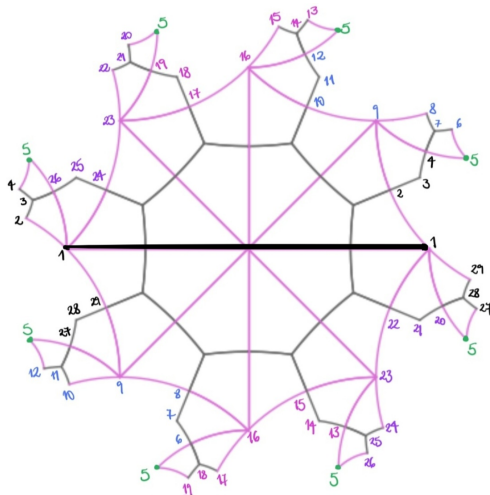


Figure: A closed geodesic on B .

Signature determines the quotient surface

Signature: $(0; 8, 3, 2)$. The surface $B/\mathrm{GL}_2(3)$ has **genus zero**.

$$\pi_1(B/\mathrm{GL}_2(3) - \{q_1, q_2, q_3\}, q) \simeq \langle x, y, z \mid x^8 = y^3 = z^2 = xyz = 1 \rangle$$

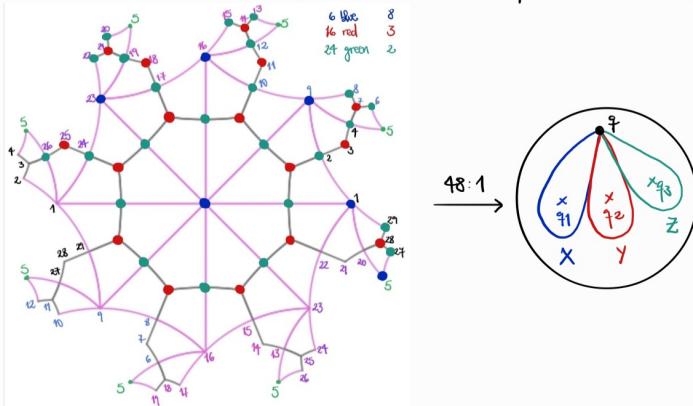


Figure: Fundamental group of the quotient.

Uniformizing the quotient

$B/GL_2(3)$ is uniformized by $\Delta \cong \langle X, Y, Z \mid X^8 = Y^3 = Z^2 = XYZ = 1 \rangle$,

$$X = \begin{pmatrix} e^{i2\pi/8} & 0 \\ 0 & 1 \end{pmatrix}$$

$$Y = \begin{pmatrix} 1 & z_0 \\ \overline{z_0} & 1 \end{pmatrix} \begin{pmatrix} e^{i2\pi/3} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -z_0 \\ -\overline{z_0} & 1 \end{pmatrix}$$

$$Z = XY.$$

The point z_0 is constrained by the hyperbolic trigonometric laws.

Fundamental quadrilateral depends on the signature

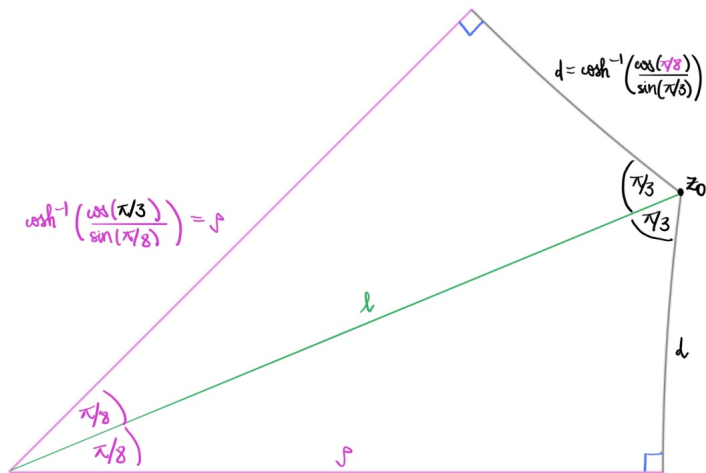


Figure: Fundamental quadrilateral for Δ . Its edges are geodesics.

Surface-kernel epimorphism relates the groups

- ▶ $B/\mathrm{GL}_2(3) \cong \mathbb{D}/\Delta$ leads to group homomorphism $E: \Delta \twoheadrightarrow \mathrm{GL}_2(3)$ with $\ker(E) = \Gamma$.
- ▶ Γ is generated by the relations needed on Δ to turn it into a finite group.
- ▶ **Generating vector:** triple $(E(X), E(Y), E(Z)) \in \mathrm{GL}_2(3)^3$.
- ▶ To find generating vectors: `with_signature_generators` (in Sage code of the thesis).

Generating vector and explicit elements

► Presentation:

$$\langle x, y, z \mid x^8 = z^2 = yzx = y^3 = (x^{-1}y^{-1}zx^{-1})^2 = zx^{-3}y^{-1}x^4y^{-1} = 1 \rangle$$

► Elements:

$$\{x^j, x^jy, x^jxyx, x^jxyx^2, x^jxyx^3, x^jxyx^3y \mid 0 \leq j \leq 7\}.$$

► Fundamental polygon for Γ : not Γ -equivalent copies of Q .

Fundamental polygon

Fundamental polygon: $P = \bigcup_{W \in M} W(Q)$,

$$M = \{X^j, X^j Y, X^j X Y X, X^j X Y X^2, X^j X Y X^3, X^j X Y X^3 Y \mid 0 \leq j \leq 7\}.$$

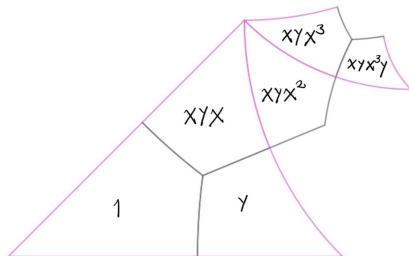
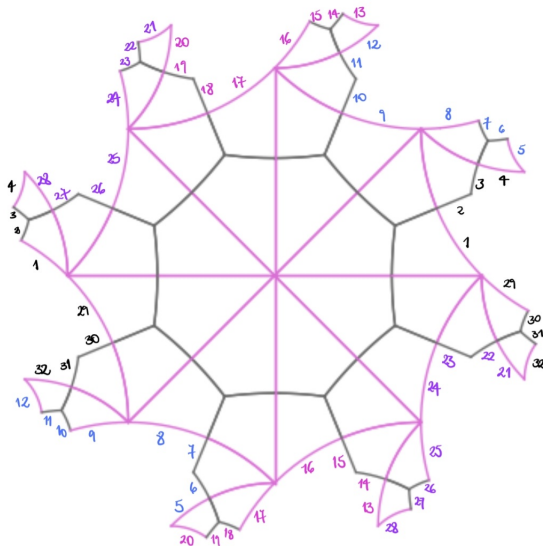


Figure: Eighth part of the polygon ($j = 0$).

Fundamental polygon



Side-pairing

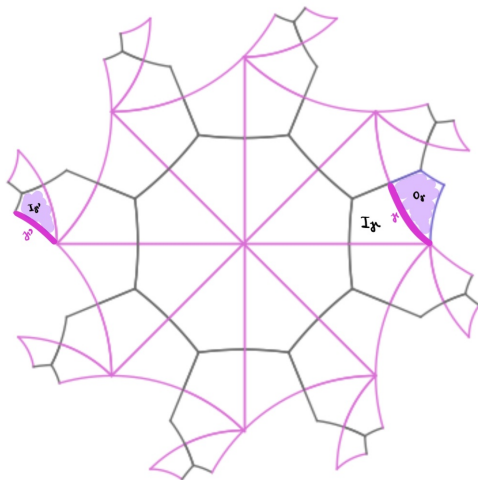


Figure: Γ -identification of boundary curves. It is performed by hand.

Boundary vertices

The boundary of P has 29 vertices and 32 edges.

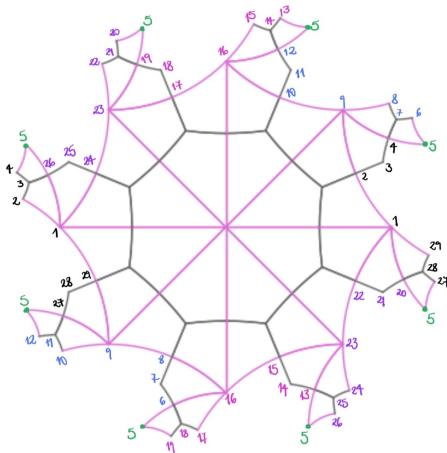


Figure: Γ -identification of the boundary vertices.

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Homology basis from relative homology

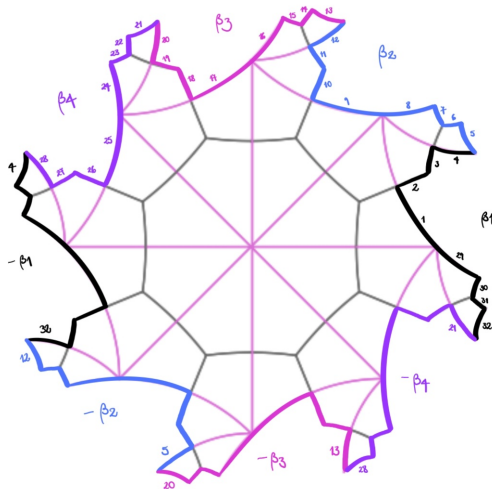


Figure: Homology basis for the Bolza surface.

Geodesic representatives (by visual inspection)

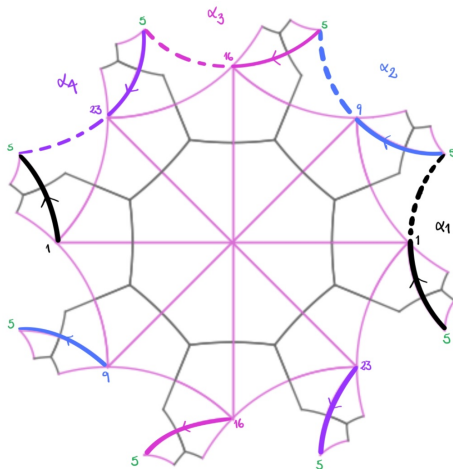


Figure: Homology basis formed by closed geodesics. Each one has length $2 \cosh^{-1}(1 + \sqrt{2})$.

- ▶ According to Schmutz Schaller (1993),
 $\text{sys}(B) = 2 \cosh^{-1}(1 + \sqrt{2}) \approx 3.0571$
- ▶ It is the biggest value of sys in $\mathcal{M}_2$.

Length of geodesic representatives provides an upper bound on sys

Action by isometries on the polygon

$GL_2(3)$ acts on the polygon by **permuting the copies of Q** .

Examples: $x \cdot Y(Q) = XY(Q)$, $y \cdot Y(Q) = XYX(Q)$.

Reductions in $GL_2(3)$ are performed using GAP.

- ▶ Induced linear action in homology: $h \mapsto HG_1(h) \in M_{4 \times 4}(\mathbb{Z})$, respect to the systolic basis $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$.
- ▶ Enough to compute **$HG_1(x)$ and $HG_1(y)$** .

Action in homology

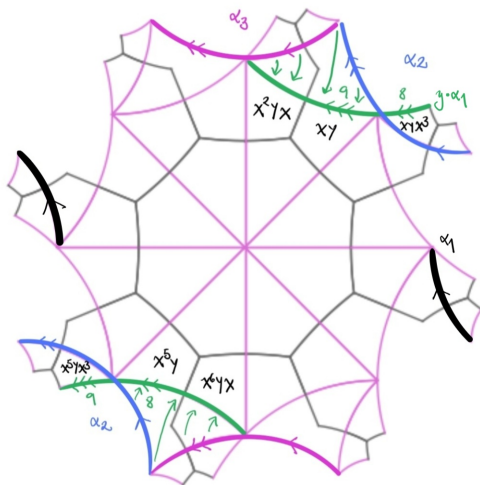


Figure: Image of α_1 under y .

$$\mathrm{HG}_1(x) = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad \mathrm{HG}_1(y) = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 1 & 0 & -1 & 1 \\ 0 & 1 & -1 & 0 \end{pmatrix}.$$

- ▶ `action_of_G` computes the 48 matrices $\mathrm{HG}_1(h)$, $h \in \mathrm{GL}_2(3)$.
- ▶ **Orbit-Stabilizer theorem:** $|\mathrm{Stab}(\alpha_1)|$ controls $|O(\alpha_1)|$.
- ▶ stabilizers: $\mathrm{Stab}(\alpha_1) = \{1, x^2yx^3, x^6yx^3, x^4\}$
- ▶ **Conclusion:** B has at least $48/4 = 12$ systoles.

The 12 systoles!

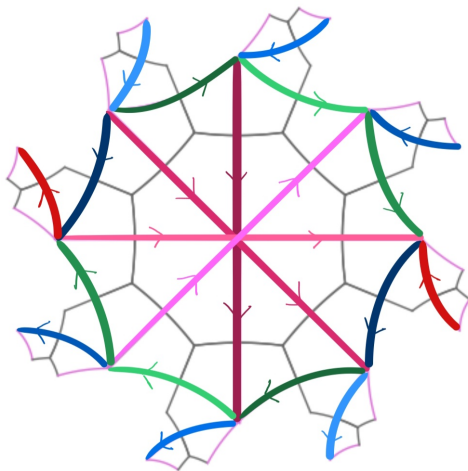


Figure: The 12 systoles of the Bolza surface.

Families on $\mathcal{M}_2$

Parametrizing polygons

Systolic bounds

Families in $\mathcal{M}_2$

Reference: Rubí Rodríguez and Anita Rojas (2024).

- ▶ $\mathcal{F}_3$ with D_6 -action with signature $(0; 3, 2, 2, 2)$,
- ▶ $\mathcal{F}_4$ with D_4 -action with signature $(0; 4, 2, 2, 2)$.

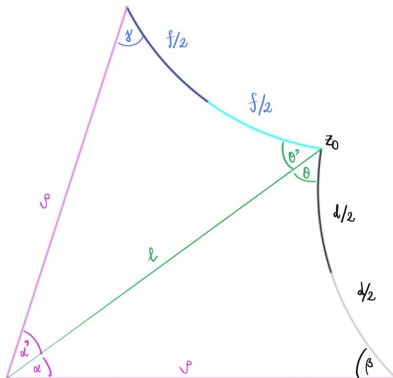


Figure: Fundamental hexagons for families with dihedral actions.

Case $f = d$

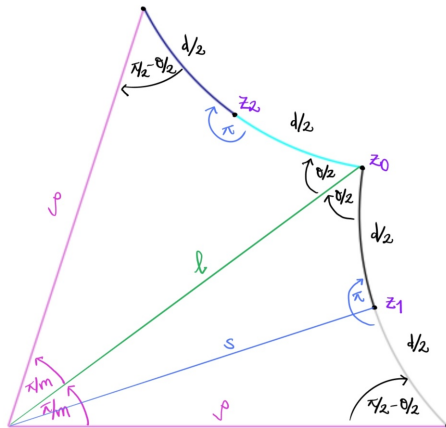
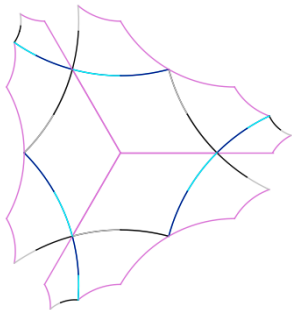
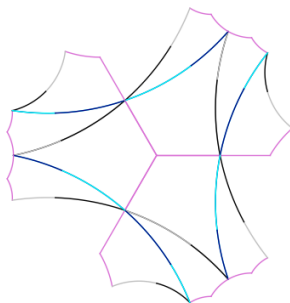


Figure: Fundamental hexagon parametrized by $\theta \in (0, \pi)$.

Fundamental polygons in $\mathcal{F}_3$



(a) $\mathcal{F}_3(\pi/2)$

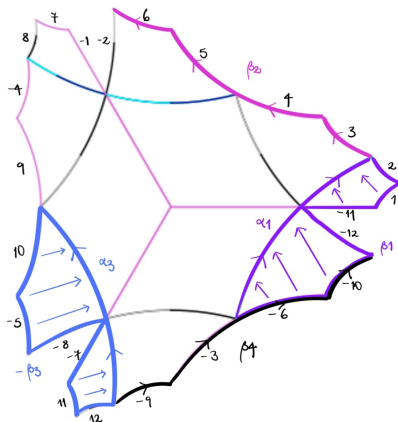


(b) $\mathcal{F}_3(\pi/8)$

Figure: Members of $\mathcal{F}_3$ with D_6 -action.

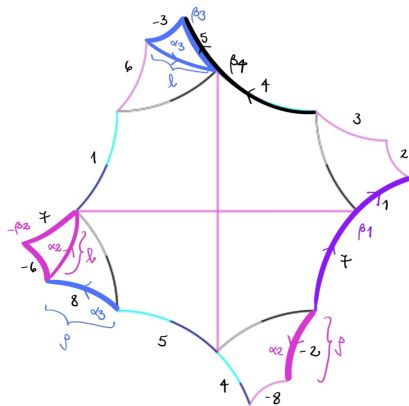
- ▶ Systolic length depends on θ ,
- ▶ Kissing number only on the group.

Systolic invariants $\mathcal{F}_3$



- ▶ α_1, α_3 : length $2d = 2 \cosh^{-1}(\sin(\theta)^{-1} + 1)$,
- ▶ β_2, β_4 : length $4\rho = 4(\cot(\pi/3) \tan(\theta/2) + \sin(\pi/3)^{-1})$,
- ▶ $\text{Kiss}(\mathcal{F}_3) \geq 12$.

Systolic invariants $\mathcal{F}_4$



- ▶ β_1, β_4 : length $2d = 2 \cosh^{-1}(\sqrt{2} \sin(\theta)^{-1} + 1)$,
- ▶ α_2, α_3 : $\ell + \rho = \cosh^{-1}(\tan(\theta/2) + \sqrt{2}) + \cosh^{-1}(\cot(\theta/2) + \sqrt{2})$
- ▶ $\text{Kiss}(\mathcal{F}_4) \geq 8$.

GAP ID	Signature	Kiss lower bound	sys upper bound ($\approx$)
(48,29)	(0;8,3,2)	12	3.0571
(12,4)	(0;3,2,2,2)	12	2.6339
(8,3)	(0;4,2,2,2)	8	3.0571

- ▶ Bounds for $\mathcal{F}_3$ and $\mathcal{F}_4$ are computed at $\theta = \pi/2$, where its *minimum* is achieved.
- ▶ The Bolza surface is contained in both families.

Conclusions and future work

Conclusions and beyond

- ▶ **Valid approach.** Recovers the systolic invariants of the Bolza surface.
- ▶ **Parametrized systolic bounds along families.**
- ▶ Promising for addressing general conjectures (Hugo Parlier, 2013):

$$\max_{S \in \mathcal{M}_g} \text{sys}(S) \leq \frac{4}{3} \log(g) + A,$$

$$\max_{S \in \mathcal{M}_g} \text{Kiss}(S) \leq Lg^{4/3}.$$

Challenges:

- ▶ adapt the fundamental polygons for other signatures,
- ▶ complete parametrization of one-dimensional families in genus two,
- ▶ automate computations in homology,
- ▶ identify geodesic representatives.

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